MIL-D-3716B
INTERIM AMENDMENT 1(USAF)
14 MARCH 1988

MILITARY SPECIFICATION

DESICCANTS, ACTIVATED FOR DYNAMIC
DEHUMIDIFICATION

This interim amendment is approved for use within the Department of the Air Force, with MIL-D-3716B dated 28 September 1987.

PAGE 1

1.2, delete and substitute:

"1.2 Classification. Desiccants shall be provided in the following types, grades, and classes as specified (see 6.2.1):

Type I - Large particle size.

Grade H - High adsorption capacity.

Class 1 - non-spherical particles.
Class 2 - spherical particles.

Grade M - Medium adsorption capacity.
Grade L - Low adsorption capacity.

Type II - Medium particle size.

Grade H - High adsorption capacity.
Grade M - Medium Adsorption capacity.

Type IV - Medium particle size impregnated with a humidity indicator.

Grade A - Extra-high adsorption capacity.
Grade H - High adsorption capacity."

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PAGE 3

3.3.2, delete

"The material shall have characteristic colors when in equilibrium with atmospheres of varying relative humidities, and shall coincide with the colors in accordance with MS16188."

and substitute

"The material shall have characteristic colors when in equilibrium with atmospheres of varying relative humidities. Grade A shall coincide with the colors in Table VII and Grade H shall coincide with the colors in MS16188.

PAGE 4

Table I, delete and substitute:

TABLE I. Minimum water vapor adsorption capacity.

Nominal relative humidity (percent)	Type IV, grade A (percent)	Types I, II, and IV, grade H (percent)	Types I and II grade M (percent)	Type I, grade L (percent)
4	2.0	-	-	-
10	5.7	5.5	14.7	3.3
20	10.5	10.0	15.6	5.0
40	21.5	19.0	16.9	7.5
60	30.0	28.0	18.2	11.7
80	33.0	33.0	19.2	15.0

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Table III, delete "IV, grade H" and substitute "IV, grades A and H"

Table IV, delete "IV, grade H" and substitute "IV, grades A and H"

3.9, delete and substitute:

"3.9 Corrosiveness. Corrosiveness shall be in accordance with 3.9.1 and 3.9.2."

Add the following paragraphs:

"3.9.1 Types I and II. When types I and II are tested as specified in 4.7.6, the test specimens of steel, brass, magnesium and aluminum in contact with the desiccant shall not be more corroded than the bare metal."

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"3.9.2 Type IV, grade A. When type IV, grade A is tested as specified in 4.7.6, test specimens of aluminum anodized per MIL-A-8625, in contact with the desiccant shall show no evidence of corrosion."

"3.9A Particle Shape, Type I, grade H, class 2. Type I, grade H, class 2 shall be in the form of spherical beads with a smooth surface having an average diameter of 0.138 to 0.141 of an inch."

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4.7.4.2, delete and substitute:

"4.7.4.2 Procedure. Thoroughly mix all material retained on the no. 18 and coarser sieves in accordance with RR-S-366, as specified in 4.7.3.2. Weigh a sample to approximately 50 grams. Place the sample on the no. 18 sieve together with five copper disks (of the size and weight of 1-cent pieces), backing the sieve with a no. 30 sieve and retaining pan. Shake the sample in the apparatus for 15 minutes. Immediately following conclusion of the specified shaker operation time, report the percentage of the desiccant passing through the no. 30 sieve which is determined by accurate weighing to within 0.2 grams. Use the sum of the individually retained weights as the total weight of the sample."

4.7.4.2.1 delete in its entirety.

PAGE 10

4.7.6, delete

"Corrosiveness type I, grades H, M, and L; type II, grades H and M.

and substitute

"Corrosiveness.

4.7.7, delete

"in accordance with MS16188"

and substitute

"in Table VII for grade A, or MS16188 for grade H."

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5.1, delete

"with MS16188"

and substitute

"with Table VII for grade A, or with MS16188 for grade H."

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6.5, delete and substitute:

"6.5 Color Standard. Information concerning the desired colors for TYPE IV are given in 6.5.1 and 6.5.2."

Add the following paragraphs and table:

"6.5.1 Type IV, Grade H. The colors given on MS16188 possibly may be found to have faded subsequent to printing. Additional information concerning the desired colors is given in table VI wherin the colors are further identified by numbers from the Munsell Book of Color, published by the Munsell Color Company Incorporated, Baltimore MD."

"6.5.2 Type IV, Grade A. Table VII gives the colors that should be clearly identified at the specified humidities for Type IV, Grade A material.

TABLE VII

Relative Humidity	Color	Approximate Munsell Color Ratings
0 - 1%	Dark Blue or Dark Bluish Green	8G4.2/4
2%	Green - Yellow	2GY5.6/6
4%	Pale Yellow	10YR7.7/11

Preparing activity:
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